A3-PS003-TC371

Federated Traffic-Flow Prediction with Communication-Cost Evaluation

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AIORI Hackathon 3

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Abstract: Problem Description

- Junction-level traffic data is rich, but pooling it centrally raises privacy, ownership and bandwidth concerns.
- Local-only models are data-starved and miss network-wide patterns; centralized models need raw-data export.
- Federated Learning keeps data on-site, yet every training round costs bandwidth, and that cost is rarely measured in traffic prediction.
- We propose CoFlow-FL: junctions collaboratively predict 15–60 min traffic flow while raw data never leaves the junction, and communication cost is tracked as a first-class metric.

How accurate?

Prediction error (MAE / RMSE / MAPE) vs centralized and local-only models.

How costly?

Megabytes exchanged per round and in total to reach a target accuracy.

How private?

Raw data stays local; updates are compressed and securely aggregated.

Background

Traffic-flow prediction

Forecasting vehicle counts at a junction 15–60 minutes ahead so authorities can act before congestion builds.

Federated Learning

Each junction trains on its own data; only model updates, never raw records, are sent for aggregation.

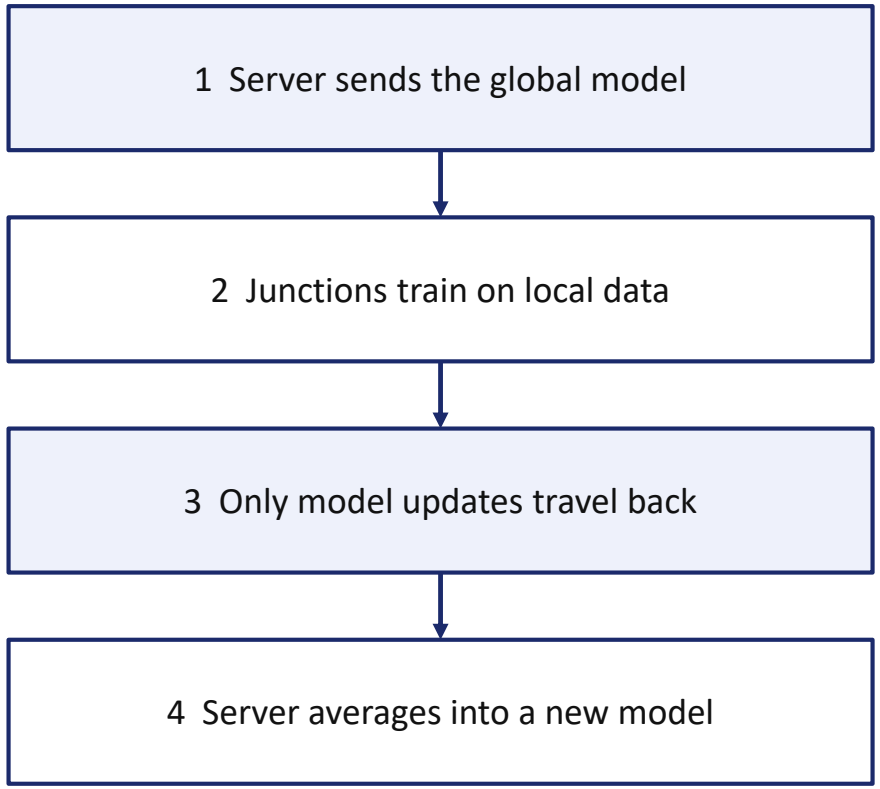
Non-IID junctions

A highway ramp and a school street behave differently, so one averaged model fits neither well.

Communication cost

Bytes up + down, per junction, per round, times rounds. This is the bottleneck on low-bandwidth city links.

One federated round (FedAvg [1])



Cost per round grows with model size × number of junctions, which motivates compression.

Literature & RFC Review

Work	What it contributes	Gap we address
FedAvg [1]	Federated averaging of locally trained models	Does not optimize bytes per round
DCRNN [2]	Graph-based traffic forecasting (METR-LA)	Needs centralized data
FL traffic prediction [3]	Federated traffic-flow forecasting with privacy focus	Communication cost not a primary evaluated metric
Deep Gradient Compression [4]	Sparsified gradients cut bandwidth sharply	Not studied for per-junction traffic models
Secure Aggregation [5]	Server sees only the sum of updates	Overhead not weighed against accuracy
Clustered FL (IFCA) [6]	Cluster clients with similar data distributions	Not tied to traffic-pattern types
Standards: RFC 8446 [9]	TLS 1.3 secures junction-to-server channels	Adopted as the transport baseline

Proposed Solution

CoFlow-FL: cost-aware, cluster-personalized federated traffic prediction

1

Local training

Each junction trains a lightweight GRU on its own sliding windows of vehicle counts.

2

Compress & gate

Top-k sparsify + 8-bit quantize the update; skip the upload if the change is insignificant.

3

Cluster

Group junctions by traffic-pattern similarity (arterial, residential, mixed) to handle non-IID data.

4

Secure aggregate

Aggregate per cluster with secure aggregation; optional noise for differential privacy.

5

Predict & measure

Forecast congestion, raise alerts, and log bytes per round in a cost meter.

What is new: accuracy-per-MB as a headline metric • update gating that skips uploads • traffic-pattern clustering • raw data never leaves the junction

Optimization Proposed

Aspect	Baseline FedAvg	Proposed CoFlow-FL
Update payload	Full float32 model sent every round	Top-k sparsified + 8-bit quantized update
Upload frequency	Every junction, every round	Gated: only significant updates are sent
Non-IID junctions	One global model for all	Cluster models + local personalized head
Privacy	Raw data local; plain updates visible to server	Secure aggregation + optional DP noise
Cost evaluation	Accuracy only	MB per round, total MB, and accuracy-per-MB
Output for authorities	Forecast values	Forecasts + congestion alerts + cost report

Improvements are design goals; we will validate them in simulation against centralized, local-only and FedAvg baselines.

Solution Architecture & Design

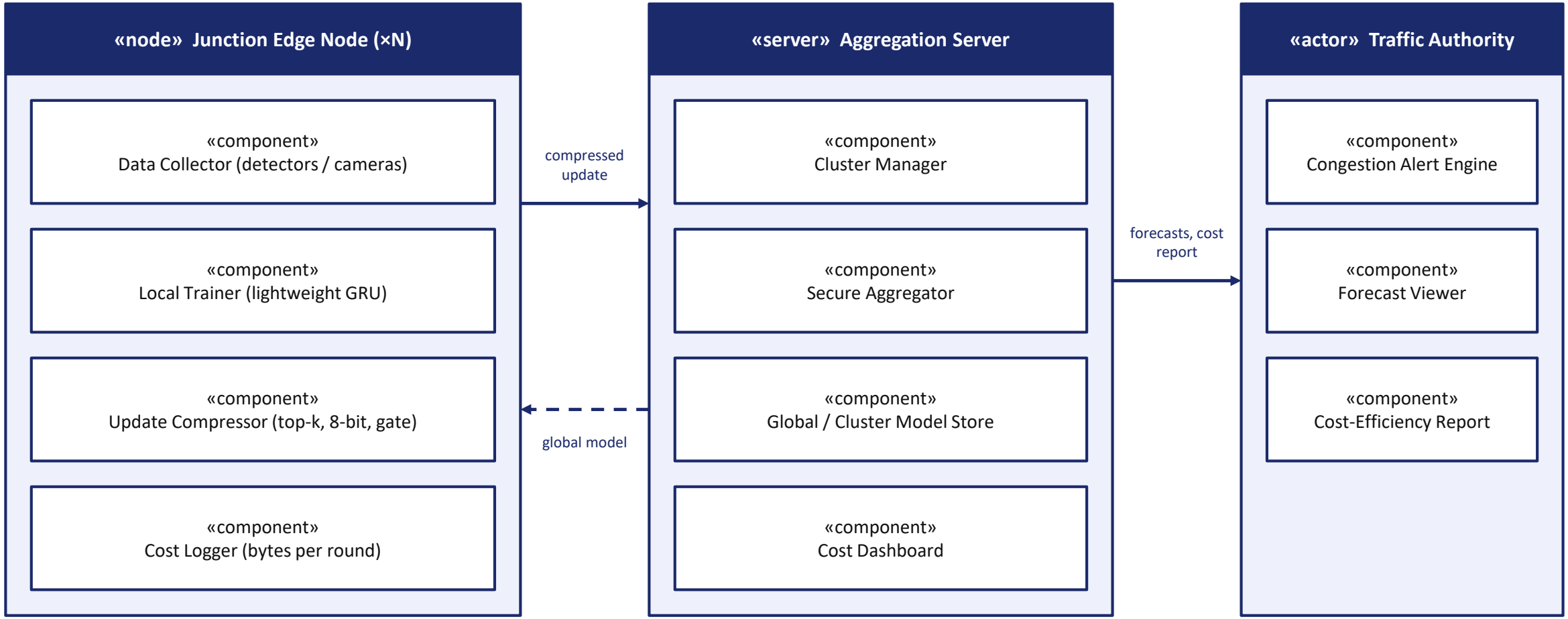
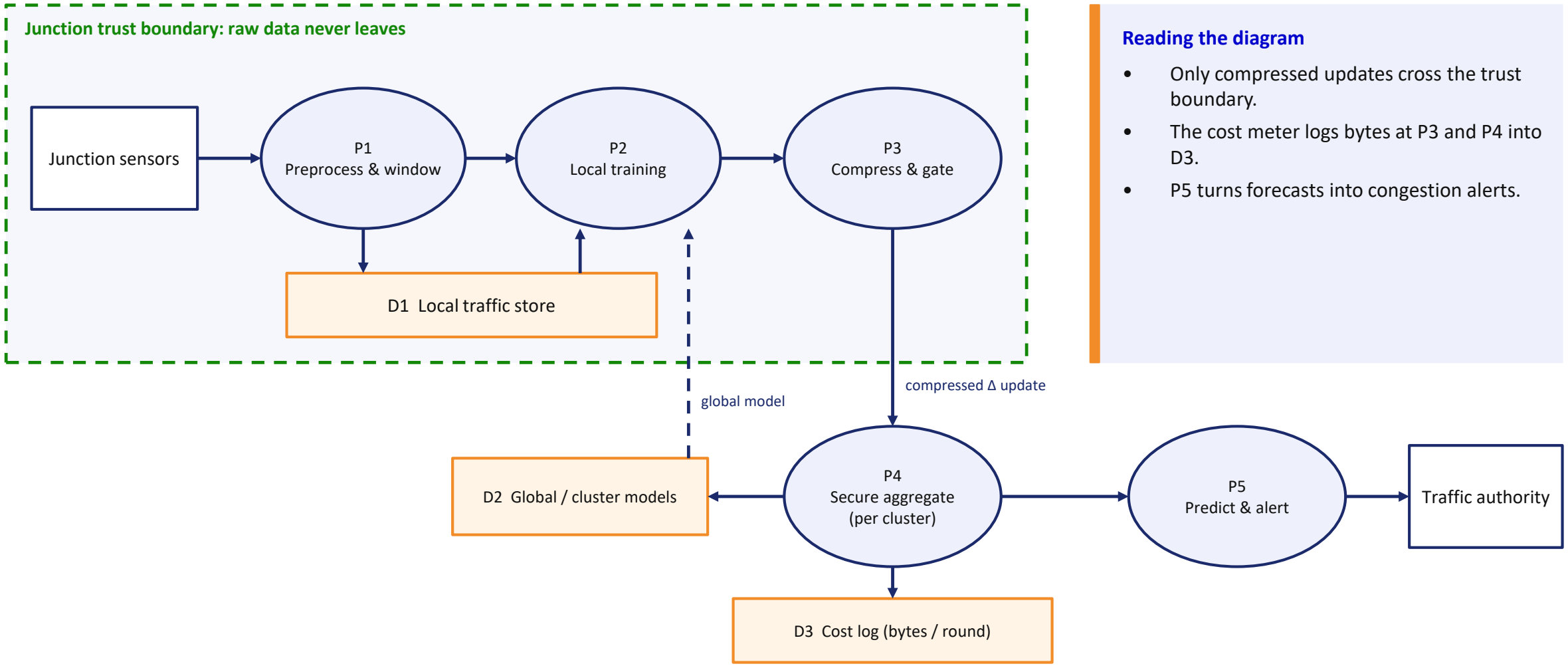


Figure: System architecture (component view). Solid arrows = upload / output, dashed = download.

Data Flow Diagram



Reading the diagram

- Only compressed updates cross the trust boundary.
- The cost meter logs bytes at P3 and P4 into D3.
- P5 turns forecasts into congestion alerts.

Deployment Timeline

1

Data & baselines

- Map METR-LA / PEMS-BAY sensors to virtual junctions
- Train centralized, local-only and FedAvg baselines

Checkpoint: Baseline RMSE and MB logged

2

Federated core

- Flower simulation with N junction clients
- Lightweight GRU local model

Checkpoint: FedAvg tracks centralized trend

3

Cost optimization

- Top-k + 8-bit quantization
- Update gating and cluster-wise aggregation

Checkpoint: Similar accuracy at fewer MB

4

Privacy & evaluation

- Secure aggregation, optional DP noise
- Privacy–accuracy–cost trade-off curves

Checkpoint: Ablation report

5

Dashboard & demo

- Congestion alerts and cost-meter dashboard
- SUMO / edge-device demo, submission pack

Checkpoint: Working end-to-end demo

Methodology: iterative and ablation-driven. Every stage ends with a measurable checkpoint (RMSE, MB per round).

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Thank You for Your Attention!

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